

EXP-013 — Document Routing

Production RAG v1 · evaluation-first · n = 20 questions / 22 evidence spans · Stage 2 frozen, only document aggregation changed

STATUS — FALSIFIED FOR RANK AGGREGATION

Not one of the three aggregation routers improved document recall@5. All four routers — including the EXP-012 control — sit at **0.875 / 17-of-20**, an identical Stage-1 ceiling of 0.850. An *exploratory* document-level embedding router reached 18/20 and recall@5 0.900, and ranked the one document no chunk-derived router could find at **8 instead of 62**.

17/20
all-required-documents routed identical for all four routers

0.875
document recall@5 unchanged by every router

0.950
oracle headroom still unrealised 19/20, not deployable

+0
document-level fusion bonus for every router

1. Why EXP-013 exists

EXP-012's oracle showed the passage layer is mostly fine — given the correct document the same retrievers reach 0.950 / 19-of-20 with zero regressions. Routing was the bottleneck: every required document reached the top 5 for only 17 of 20 questions. The router it used keeps a document's single best chunk and discards the rest, so EXP-013 tests whether the discarded evidence routes better.

2. Frozen, and why everything stays in the rank domain

Stage 2 is untouched: raw query, full-corpus BM25 plus transformer cosine restricted to routed documents, passage RRF k=60, top 10, statistics never recomputed locally. Chunks, model (fingerprint `bd95feaeacf98559`), embeddings, anchors and questions all unchanged; `top_documents = 5` throughout.

BM25 and cosine scores are never combined directly — a test asserts the router module contains no such expression. Mixing their scales, or normalising them against 20 questions and tuning a weight, would fit the evaluation set rather than measure on it. `k=60`, support 5 and vote depth 50 were preregistered and not tuned.

3. Reproduction gates

GLOBAL control (0.775 / 15-of-20), A_MAX (0.725 / 14-of-20) and ORACLE (0.950 / 19-of-20) all reproduced exactly — **PASS**.

4. Routing quality — the primary metric

router	@1	@3	@5	@10	all-routed	ceiling	bonus	mean pool
A_MAX — best chunk (EXP-012 control)	0.525	0.800	0.875	0.950	17/20	0.850	+0	900
B_RANK_SUM — reciprocal-rank support	0.475	0.825	0.875	0.900	17/20	0.850	+0	1745
C_TOPK_VOTE — breadth of support	0.475	0.825	0.875	0.925	17/20	0.850	+0	1446
D_MAX_SUPPORT — both, rank-domain fusion	0.575	0.875	0.875	0.950	17/20	0.850	+0	1348

router	@1	@3	@5	@10	all-routed	ceiling	bonus	mean pool
E_DOC_EMBED — exploratory, fused	0.675	0.875	0.900	0.950	18/20	0.900	—	—

Aggregation helps at *shallow* depth — D_MAX_SUPPORT lifts recall@1 from 0.525 to 0.575 and recall@3 from 0.800 to 0.875 — and every router has converged by depth 5. The routers behave exactly as designed on the motivating example (a document with support at ranks 2, 7, 11, 18 beats one owning only rank 1); the corpus simply does not reward it where it matters.

5. End-to-end

cell	macro R	full	spans@10	doc R	MRR	absent@300	oracle gap
GLOBAL control	0.775	15/20	17/22	0.925	0.449	2	0.175
A_MAX	0.725	14/20	16/22	0.875	0.421	4	0.225
B_RANK_SUM	0.775	15/20	17/22	0.875	0.414	4	0.175
C_TOPK_VOTE	0.775	15/20	17/22	0.875	0.412	4	0.175
D_MAX_SUPPORT	0.775	15/20	17/22	0.875	0.402	4	0.175
ORACLE — not deployable	0.950	19/20	21/22	1.000	0.646	0	0.000

Aggregation **repairs** the EXP-012 hierarchy regression — B, C and D each recover to global parity — but **none exceeds global**. Against the EXP-012 hierarchy each scores net +1: rescuing **OA-004** and AN-011, losing **AN-008**. Against global the exchange nets to zero.

OA-004 is the aggregation success. Its document has 5 BM25 and 3 transformer chunks inside the top 30 — breadth A_MAX discarded. Document rank moves 6 → 2 and its evidence lands at rank 2–4. **AN-008 is the cost:** the same breadth preference demotes it out of the routed set.

6. Why the ceiling never moves

router	cases missing a required document
A_MAX	AN-001, AN-012, OA-004
B_RANK_SUM	AN-001, AN-008, AN-012
C_TOPK_VOTE	AN-001, AN-008, AN-012
D_MAX_SUPPORT	AN-001, AN-008, AN-012

AN-001 — THE CASE THAT EXPLAINS THE WHOLE RESULT

Its document contributes **one** chunk anywhere in 300 BM25 results (best rank 19), and the transformer **never retrieves it at all**. Document rank 62– 72 under every router. Yet handed the document, its evidence ranks **1**.

There is nothing for an aggregation rule to aggregate. **The limitation is the input, not the arithmetic.**

AN-012 is multi-hop: one document is easy (transformer chunk rank 2), the other has transformer rank 63 and lands at document rank 9–13 — just outside the top 5 under every router. Partially routed is, for a multi-hop question, the same as failing.

7. Complementarity — document-level fusion does no work

router	BM25 alone	transformer alone	best component	fused	bonus
A_MAX	15/20	17/20	17	17	+0
B_RANK_SUM	16/20	17/20	17	17	+0
C_TOPK_VOTE	16/20	17/20	17	17	+0
D_MAX_SUPPORT	15/20	17/20	17	17	+0

The transformer's document ranking alone already achieves 17/20; BM25 adds nothing and fusion adds nothing over the better component. That is a different picture from passage retrieval, where the raw fusion bonus is +4 cases. **Retriever complementarity is a passage-level phenomenon here, not a document-level one.**

EXPLORATORY — A DOCUMENT-LEVEL REPRESENTATION

Run only after A–D were frozen: the mean of each document's already-stored normalised chunk vectors, renormalised, ranked by cosine. No re-chunking, no training, no tuning.

Alone it reaches recall@10 of **1.000** — every expected document inside its top 10, which no chunk-derived ranking achieves — though only 16/20 at depth 5. **Fused with chunk routing** it reaches **18/20** and recall@5 **0.900**, the first configuration in EXP-013 to exceed 17/20. And **AN-001's document moves from rank 62 to rank 8**.

This is one exploratory run at n = 20 and it is not a result. One case is 5 percentage points, and the same fusion that gains AN-012 gives back OA-004. A direction with measured support, not a finding.

8. AN-003 — unchanged, as expected

Routed correctly by every router (document rank 3–4) and still absent@300, exactly as EXP-012 predicted. Its document was never the problem; its oracle rank remains 29 inside its own 141-chunk document. Nothing here was modified to target it.

9. Reranker gate

cell	1–10	11–30	31–50	51–100	101–300	absent	c@30	c@50	c@100
GLOBAL	17	1	1	1	0	2	0.818	0.864	0.909
best router (D)	17	0	1	0	0	4	0.773	0.818	0.818
ORACLE	21	1	0	0	0	0	1.000	1.000	1.000

Routing improvements did not raise the reranker ceiling. Under the best router it is **0.818**, still *below* the global control's 0.909 — hierarchy's exclusions remove candidates a reranker could otherwise have reordered. Only oracle routing reaches 1.000, and it does so at a pool of just 30.

10. Limitations

- n = 20 / 22 spans; one case is 5 percentage points. No significance claims.
- Three aggregation rules were tested, not the space of aggregation. Their agreement at recall@5 is suggestive, not exhaustive.

- Router E is exploratory, unreplicated, and its gain is a single case bought at the cost of another.
- Mean-pooled document vectors are the crudest possible document representation — chosen because they need no tuning, not because they are good.
- **EXP-NULL remains BLOCKED** — still no measured no-retrieval floor.

11. Did better document aggregation solve routing?

No. Three preregistered rules, all landing on exactly the same recall@5 and the same 17/20. The Stage-1 ceiling did not move by a single case. This is preregistered **Outcome D**: chunk-ranking-derived routing is itself inadequate, and AN-001 shows why in one line. **The routers were not the problem, the input was.**

12. What the measurements justify next

1. **Stop inventing aggregation formulas.** Three rules, one ceiling.
2. **A document-level retrieval representation is the justified next experiment** — the exploratory router already reaches recall@10 = 1.000 alone and moves AN-001 from 62 to 8. It deserves a proper preregistered experiment, not promotion off one run.
3. **Do not deploy hierarchy.** No router beat global, and hierarchy still lowers the reranker ceiling.
4. **A reranker remains premature** — behind current routing its ceiling is worse than global's. Routing first, reranking second.
5. **Complementarity is passage-level, not document-level.** A future document retriever should be judged on whether it adds anything the transformer does not already have.

Generated from experiments/EXP-013/* .json by scripts/build_exp013_pdf.py. Git commit d94135811e4d. Config hash 551bb8b41d22f795. Router version 1.0; k=60, support 5, vote depth 50, document RRF k=60 — all preregistered, none tuned. Snapshot snap_689e336380a054d8039dc35b2c09cd0a, chunk set cs_v1_control, top_documents 5. Stage 2 frozen. Raw provider documentation is not redistributed.